

Lab Manual

Topic: Control Instructions — Conditional Statements

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Background

Control instructions direct the flow of a program. Without them, code executes line by line from top to bottom — but real programs need to make decisions. **Conditional statements** allow a program to choose between different paths of execution based on whether a condition evaluates to `True` or `False`.

Key Concepts

Boolean Expressions

A condition is any expression that resolves to `True` or `False`. Common comparison operators include:

- `==` (equal to), `!=` (not equal to)
- `<`, `>`, `<=`, `>=`

Logical Operators

Multiple conditions can be combined using:

- `and` — both conditions must be true
- `or` — at least one condition must be true
- `not` — inverts the condition

Types of Conditional Constructs

Construct	Purpose
<code>if</code>	Execute a block only if condition is <code>True</code>
<code>if-else</code>	Choose between two blocks
<code>if-elif-else</code>	Choose among multiple alternatives
Nested <code>if</code>	A conditional inside another conditional

Note: Python uses **indentation** (4 spaces) to define code blocks — there are no braces `{}` like in C/Java.

Lab Tasks (With Solutions)

Task 1 — Simple if Statement: Check Positive Number

Objective: Understand the basic `if` statement.

Background: The simplest conditional checks one condition. If the condition is `True`, the indented block runs; otherwise, Python skips it entirely.

Problem: Write a program that takes an integer input from the user and prints "Positive number" only if the number is greater than zero.

Solution:

```
num = int(input("Enter a number: "))
if num > 0:
    print("Positive number")
```

Sample Output:

```
Enter a number: 7
Positive number
```

Explanation:

- `int(input(...))` reads user input and converts it to an integer.
 - `if num > 0:` checks whether the value exceeds zero.
 - If the condition is `False` (e.g., input is `-3`), nothing is printed.
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Task 2 — if-else Statement: Even or Odd

Objective: Use `if-else` to handle two mutually exclusive outcomes.

Background: When exactly two outcomes are possible, `if-else` is the cleanest choice. The `else` block runs whenever the `if` condition is `False`.

Problem: Ask the user for an integer. Print "Even" if it is divisible by 2, otherwise print "Odd".

Solution:

```
num = int(input("Enter an integer: "))
if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Sample Output:

```
Enter an integer: 4
Even

Enter an integer: 9
Odd
```

Explanation:

- The modulo operator `%` gives the remainder of division.
 - `num % 2 == 0` is `True` when there is no remainder, i.e., the number is even.
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Task 3 — `if-elif-else`: Grade Classification

Objective: Handle multiple ranges of values using `elif`.

Background: `elif` (short for "else if") lets you chain additional conditions. Python checks them in order and executes only the first block whose condition is `True`. The `else` at the end acts as a catch-all.

Problem: Read a numeric score (0–100) and print the letter grade according to the table below.

Score Range	Grade
90 – 100	A
80 – 89	B
70 – 79	C
60 – 69	D
Below 60	F

Solution:

```
score = int(input("Enter your score (0-100): "))  
  
if score >= 90:  
    print("Grade: A")  
elif score >= 80:  
    print("Grade: B")  
elif score >= 70:  
    print("Grade: C")  
elif score >= 60:  
    print("Grade: D")  
else:  
    print("Grade: F")
```

Sample Output:

```
Enter your score (0-100): 85  
Grade: B
```

Explanation:

Because Python stops at the first `True` condition, we only need `>=` comparisons — we don't need to write `80 <= score < 90`.

Task 4 — Nested `if`: Loan Eligibility

Objective: Use nested conditionals to evaluate multiple layered criteria.

Background: Sometimes a decision depends on a hierarchy of conditions — you only ask the

second question if the first condition is already met. This is done by placing an `if` statement inside another `if` block.

Problem: A bank approves a loan if:

1. The applicant's age is at least 21, **and**
2. Their monthly salary is at least 30,000.

Print "Loan Approved" or an appropriate rejection reason.

Solution:

```
age = int(input("Enter your age: "))
salary = float(input("Enter your monthly salary: "))

if age >= 21:
    if salary >= 30000:
        print("Loan Approved")
    else:
        print("Loan Rejected: Salary too low")
else:
    print("Loan Rejected: Applicant must be at least 21 years old")
```

Sample Output:

```
Enter your age: 25
Enter your monthly salary: 25000
Loan Rejected: Salary too low
```

Explanation:

The outer `if` gates access to salary checking. This structure gives a precise reason for each rejection, which `and` in a single condition cannot do alone.

Task 5 — Logical Operators: Leap Year Checker

Objective: Combine multiple conditions using `and` / `or`.

Background: A year is a **leap year** if:

- It is divisible by 4, **and**
- It is **not** divisible by 100, **unless** it is also divisible by 400.

This rule requires combining `and`, `or`, and `not` (or their equivalents).

Problem: Take a year as input and print whether it is a leap year or not.

Solution:

```
year = int(input("Enter a year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print(f"{year} is a Leap Year")
else:
    print(f"{year} is NOT a Leap Year")
```

Sample Output:

```
Enter a year: 2000
2000 is a Leap Year

Enter a year: 1900
1900 is NOT a Leap Year
```

Explanation:

- $2000 \% 400 == 0 \rightarrow \text{True} \rightarrow \text{Leap Year}.$
 - $1900 \% 4 == 0 \rightarrow \text{True}$, but $1900 \% 100 == 0 \rightarrow$ the first part is False, and $1900 \% 400 != 0 \rightarrow$ overall False $\rightarrow$ Not a leap year.
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Student Practice Problems

Instructions: Solve the following problems on your own. Write, run, and test your code. No solutions are provided here — refer to your solution sheet only after attempting each problem.

Problem 1.

Write a program that takes a number as input and prints:

- "Positive" if the number is greater than 0
 - "Negative" if the number is less than 0
 - "Zero" if the number equals 0
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Problem 2.

A shop offers a 15% discount if a customer's bill exceeds Rs. 5,000, and a 5% discount if the bill is between Rs. 2,000 and Rs. 5,000 (inclusive). No discount is given otherwise. Take the bill amount as input and print the final amount after discount.

Problem 3.

Take three numbers as input and print the largest among them using `if-elif-else`.

Problem 4.

Write a program that checks whether a given character (input from the user) is a vowel (a, e, i, o, u) or a consonant. Assume the input is always a single lowercase letter.

Problem 5.

A traffic light system uses colors: "red", "yellow", "green". Take the color as input and print:

- "Stop" for red

- "Get Ready" for yellow
 - "Go" for green
 - "Invalid color" for anything else
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Problem 6.

Write a program that determines if a triangle is **equilateral**, **isosceles**, or **scalene** based on three side lengths entered by the user.

Problem 7.

A cinema charges:

- Rs. 200 for children (age < 12)
- Rs. 350 for students (age 12–25 with a student card)
- Rs. 500 for adults

Take age and whether the person has a student card (yes / no) as inputs. Print the ticket price.

Problem 8.

Write a program that reads a year and a month number (1–12) and prints the number of days in that month. Account for leap years in February.

Problem 9.

A university admission system accepts students if:

- Their FSc marks are at least 60%, **and**
- Their entry test score is at least 50 out of 100.

If only one condition is met, print which requirement was not fulfilled. If neither is met, print both.

Problem 10.

Write a simple calculator. Take two numbers and an operator (+, -, *, /) as input. Perform the corresponding operation and print the result. Handle division by zero with an appropriate message.

Problem 11.

Write a program that takes a person's weight (kg) and height (m) as input, calculates their BMI ($\text{BMI} = \text{weight} / \text{height}^2$), and classifies it as:

- Underweight: $\text{BMI} < 18.5$
- Normal: $18.5 \leq \text{BMI} < 25$
- Overweight: $25 \leq \text{BMI} < 30$

- Obese: BMI ≥ 30
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Problem 12.

Write a number guessing hint program. Take a secret number (1–100) and a guessed number as input. Print:

- "Too High" if the guess is more than the secret number
 - "Too Low" if the guess is less
 - "Correct!" if they match
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— *End of Lab Manual* —